

## **WG21 June 2023 Hybrid meeting**

### **Record of Discussion**

ISO/IEC JTC1 SC22 WG21 P2931— 2023-28-06

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Chair: John Spicer

12-17 June 2023, Varna, Bulgaria

## **1. Opening activities**

John Spicer opens the meeting at 09:03 AM GMT+3

### **1.1 Opening comments, welcome from host (INCITS C++)**

Vassil Vassilev welcomes the group and presents information about food options, social events during the meeting, and organizational contacts.

A word from the sponsors

Vassil Keremidchiev from Chaos welcomes the group.

Stanimir Lukanov from VMware welcomes the group.

Thank you to the sponsors and meeting organizers.

### **1.2 Meeting guidelines**

John Spicer presents.

Please speak into the microphone so people participating over Zoom can hear. Please introduce yourself when speaking.

Meetings are not public, we want everyone to be able to speak freely. Please refrain from live tweeting, blogging, taking photos of other people's screens or recording the meetings. You're allowed to take screenshots of presentations for your personal use.

Agenda is on the wiki.

Every participant is responsible for understanding and abiding by the following:

[The INCITS Antitrust Guidelines](#) (INCITS C++)

[The INCITS Patent Policy](#) (INCITS C++)

[The ISO Code of Conduct](#)

[The INCITS Code of Conduct \(INCITS C++\)](#)

[The IEC Code of Conduct](#)

[The WG21 Practices and Procedures, and Code of Conduct](#)

John Spicer presents the slides. They are also linked in the agenda.

If you have any questions or concerns about CoC issues, please approach a committee officer or a NB representative and bring it to their attention.

## 1.3 Membership, voting rights, and procedures for the meeting

John Spicer presents voting rights.

If you are representing an organization that is considering formally joining INCITS/C++, or your organization is already a member and you wish to change your voting status, please inform Barry Hedquist, Nevin Liber or me (John Spicer).

On Saturday we take WG21 plenary polls. Rules are the same for US and other NBs. Every person registered in the global directory gets one vote. If you have questions about your voting status, please contact one of the officers.

Everyone can vote in subgroups. Please see best practice for voting.

Nevin Liber presents. Please sign the attendance list. It has been emailed to the reflector, it is available on the Mattermost, and it can be found on the wiki.

Jonathan Caves : Do we need a google account to access it?

Nevin Liber : you should be able to just click on the link.

John Spicer : please speak into the microphone so people on Zoom can hear you.

Nina Ranns : please introduce yourself before speaking.

Nevin Liber : if you need to upload papers to the paper system, but don't have a log in, let me know. The next official mailing will be on July 15th.

John Spicer presents voting procedures during the plenary.

## 1.4 INCITS/C++ Social Media Information (Linkedin, Twitter, Facebook)

Click [here](#) for the INCITS social media landing page.

## 1.5 Introductions

New members introduce themselves.  
John Spicer welcomes new members.

Herb Sutter presents. As of now, we have 18 national bodies represented at the meeting. From the chat, I see that we expect one or two more to join during the week.

## 1.6 Agenda review and approval (INCITS/C++ motion, WG21 poll)

John Spicer presents the agenda.

The primary goals of this meeting will be work on C++26 features.  
Additional, lower-priority goal includes subgroup work that may target versions after C++26.

INCITS/C++ business.

WG21 motion to approve the meeting agenda.  
The motion is unanimously approved by WG21.

## 1.7 Editor's reports, approval of working drafts

| Document                | Editor's report       |
|-------------------------|-----------------------|
| C++ 26 Working Draft    | <a href="#">N4951</a> |
| Library Fundamentals V3 | <a href="#">N4949</a> |

WG21 motion to approve the above.  
The motion is unanimously approved by WG21.

## 1.8 Approval of the minutes of the previous meetings (INCITS C++ motion, WG21 poll)

| Meeting             | Minutes                            |
|---------------------|------------------------------------|
| WG21 Issaquah       | <a href="#">N4943</a>              |
| INCITS C++ Issaquah | <a href="#">pl22.16-2023-00012</a> |

INCITS/C++ business.

WG21 motion to approve the above.

The motion is unanimously approved by WG21.

## 2. Liaison reports, and WG21 study group reports (see pre-meeting WG21 telecon minutes)

No discussion.

## 3. WG progress reports (Core, Evolution, Library, Library Evolution; see pre-meeting WG21 telecon minutes)

No discussion.

## 4. New business requiring action by the committee

No discussion

## 5. Organize working groups and study groups, establish working procedures

Jens Maurer presents.

Thank you to those who brought audio setup gear and all the volunteers who helped set it up. This helps the virtual attendance.

Jens Maurer presents the time schedule.

Jens presents the subgroup room assignment.

Jens presents the evening sessions.

John Spicer : heroic effort from Jens in organizing these meetings. Thank you.

## 6. Subgroup sessions

John Spicer presents. The subgroup chairs must arrange for any proposals to be written up in the form of a motion, and made available by 8:00 PM Friday on the straw polls page together with associated papers. Groups are encouraged to make those papers and polls available as soon as possible during the week so people can have time to review them.

Jens Maurer : on the core wiki page, there is a link to the core poll staging area. It will be moved to the main poll page on Friday. Same for library. Respective chairs will make an effort to update the staging area.

John Spicer Presents. There is general information on the main Varna wiki page that will be useful to new members, including information about the communication channels we use.

If you have questions or concerns regarding polls/papers, we want to know as soon as possible. Raise them with subgroup chairs, national body, on the reflector or through the authors.

## 7. Review of the meeting

Subgroup status and progress reports. Presentation and discussion of proposals to be considered for consensus adoption by full WG21.

### SG1: Concurrency (Giroux)

No report.

### SG2: Modules (Stone)

Currently dormant.

### SG4: Networking (Snyder)

SG4 has no open papers and has not met since Issaquah.

### SG5: Transactional memory (Boehm/Maurer)

Currently dormant.

### SG6: Numerics (Kretz/Lippincott/McFarlane)

SG6 met for one session to look at P2746R2 "Deprecate and Replace Fenv Rounding Modes" and P2901R0 "Extending linear algebra support to batched operations". P2746R2 will come back after receiving feedback. P2901R0 was acknowledged by SG6 and forwarded as having no SG6 relevant issues that need work.

## SG7: Compile-time programming (Dusikova/Vandevoorde)

SG7 met and had only one paper D2911R0 "Python Bindings with Value-Based Reflection" which was a study / usage report of implementation of current status quo.

## SG9: Ranges (Levi/Carter)

The Ranges Study Group had a hybrid meeting on Monday. We had a productive meeting in which we reviewed:

P2846R0: `size_hint`: Eagerly reserving memory for not-quite-sized lazy ranges

P2728R4: Unicode in the Library, Part 1: UTF Transcoding

P2542R2: `views::concat`

The first paper (P2846R0) was forwarded to LEWG for C++26, with a design question (whether or not optimization should be mandatory), and with a name change recommendation. The third paper had all the fixes and tests previously requested by SG9 and was forwarded to LEWG (P2542R2: `views::concat`) The second paper (P2728R4) got feedback from the group but will require more work, and will be discussed in SG9's future meetings. A few additional "Ranges" papers are in the queue of LEWG and will be seen there in the next few weeks.

Thank you to all our participants, and special thanks to the ones who had to overcome the time zone challenges.

## SG10: Feature test (Revzin/Wakely)

No report

## SG12: Undefined and unspecified behavior (Ballman/Wakely)

Currently dormant.

## SG14: Games & low latency (Wong)

SG14 did not meet but will continue monthly meetings rotating between Games, Low Latency, Embedded. Recently, we have been working through a paper by Patrice Roy collected from the games domain.

## SG19: Machine Learning (Wong/Reverdy)

SG19 did not meet but will continue rotating between Statistics, Graph, and Matrix. There are now 2 Statistics papers and we anticipate Graph will be voted out soon.

## SG15: Tooling (Spencer/Boeckel)

No report.

## SG18: LEWG Incubator (Baker/Liber)

SG18 did not meet in Varna.

## SG16: Unicode (Honermann/Brett)

SG16 did not meet this week, but intends to continue its twice monthly telecon cadence for the foreseeable future. We continue to have no shortage of work ahead of us.

## SG17: EWG Incubator (Keane/TBD)

Saw 6 papers over 2 afternoons

- P2889R0 Distributed Arrays
  - Motivated, but needs work
- P2893R0 Variadic Friends
  - Forwarded to EWG
- P2785R1 Relocating PRValues
  - No Consensus, but author encouraged to discuss with other relocatable authors
- Provided feedback without vote to 3 other papers:
  - P2665R0 Allow calling overload sets containing T, const T&
  - P2666R0 Last use optimization
  - P2668R0 Role based parameter passing

## SG20: Education (van Winkel)

SG20 met in Varna on Wednesday, there were 7 attendees (tough competition from SG23 - safety & Security). We had a good discussion of how to move forward with our teaching guidelines, how to get more contributions, and how we can improve on the presentation of the document. We also developed ideas around our advising function towards WG21 in general, which we will work out in the coming months.

## SG21: Contracts (Spicer/Doumler)

SG21 met for 1.5 days in Varna (Tue AM, Thu all day). We are working on a Contracts MVP proposal that can be forwarded to EWG/LEWG for C++26. We discussed three papers:

- P2877R0 "Contract Build Modes, Semantics, and Implementation Strategies" by Joshua Berne and Tom Honermann
- P2811R6 "Contract-violation handlers" by Joshua Berne
- P2853R0 "Proposal of std::contract\_violation" by Andrew Tomazos

We had consensus on the first two papers, and consensus against the third. As a consequence, we removed the notion of build modes from the Contracts MVP. Every contract annotation now has one of the following three semantics: ignore, observe, enforce, and it is implementation-defined which one you get. Further, we now have a consensus design for contract-violation handling.

Between now and Kona our main focus will be on syntax. We will also look at papers dealing with various other aspects of the design that are still missing from the Contracts MVP (for example, how should contracts behave during constant evaluation? how should they interact with lambdas? etc.)

## SG22: C/C++ Liason (TBD)

Did not meet. SG22 usually meets via telecon to allow WG14 members to attend, but that hasn't happened for a while. Applications for the SG22 chair are warmly accepted.

## SG23: Safety/Security (Orr/Craig)

We met for half a day and had three subjects for discussion:

P2771R1 towards memory safety in C++

Following the presentation we took a poll which had consensus against further work in this direction.

P2878R1 Reference checking

Following the presentation we took a poll which had consensus against further work in this direction.

Presentation on further work on profiles

Bjarne gave an update on the profiles approach and asked for volunteers to assist with further work.

No polls were taken.

## ABI Group (Vandevoorde)

No report.

## Admin (Liber)

If you haven't marked the attendance sheet, please do so now.

Admin did not meet this week. The next mailing is scheduled for July 15th.

## Evolution (Bastien)

This week, EWG saw all issues and papers which had a presenter (in person or online), and were not marked as needing revision. This amounts to 3 issues, and 30 papers being discussed, at the end of which we have 98 papers needing revision. Some of the papers

marked as needing revision have been in this status for a while, the chair will follow up with authors to ensure papers either make forward progress or terminate.

EWG also hosted an open discussion regarding the Lakos rule ([P2861](#), [P2834](#), [P2831](#), [P2837](#)), now that we've shared views we expect to host a joint session of EWG/LEWG/SG21 in Kona. We will attempt to determine group policies such as those for noexcept in this session.

We are starting on C++26, and already a few language feature papers were voted into C++26 this week:

- [P2621R2](#) UB? In My Lexer?
- [P1854R4](#) Making non-encodable string literals ill-formed
- [P2361R6](#) Unevaluated strings
- [P2558R2](#) Add @, \$, and ` to the basic character set
- [P2738R1](#) constexpr cast from void\*: towards constexpr type-erasure
- [P2552R3](#) On the ignorability of standard attributes
- [P2752R3](#) Static storage for braced initializers
- [P2741R3](#) user-generated static\_assert messages
- [P2169R4](#) A nice placeholder with no name

The 9 EWG papers which had no presenters this week are:

- [P2893](#) Variadic Friends
- [P2784](#) Not halting the program after detected contract violation
- [P2633](#) thread\_local\_inherit: Enhancing thread-local storage
- [P2624](#) Make operations on bools more portable
- [P2607](#) Let alignas specify minimum alignment
- [P2355](#) Postfix fold expressions
- [P2191](#) Modules: ADL & GMFs do not play together well (anymore)
- [P1046](#) Automatically Generate More Operators
- [P1203](#) modular main()

The chair will see if presenters can be found for the next meeting.

After the week's activities, we are left with these issues and papers which are now ready to be seen:

- [CWG2726](#) Alternative tokens appearing as attribute-tokens
- [CWG1699](#) Does befriending a class befriend its friends?
- [CWG2669](#) Lifetime extension for aggregate initialization
- [P2865](#) Remove Deprecated Array Comparisons from C++26
- [P2795](#) Correct and incorrect code, and "erroneous behaviour"
- [P0342](#) pessimize\_hint
- [P2662](#) Pack Indexing

They will be seen in Kona.

Given the light open workload, EWG will not host any telecons between now and Kona.

## Library Evolution (Adelstein Lelbach/Fracassi/Craig)

At 2023-06 Varna

We planned to focus on big papers at F2F meetings, and we delivered.

Advanced to Library:

- P1928R4 SIMD
- P0876R13 Fibers
- P0843R6 inplace\_vector

Progression on:

- P0447R22 hive
- P0260R5 Concurrent Queues
- P1030R5 path\_view
- P2781R2 constexpr\_v

For hive paper, we need to refine the wording. Anyone who has interest in hive, please reach out to me and the author to get involved. If you feel it may not belong in the standard library, we would encourage you to prepare a paper with the objections so we can discuss it.

Future Plans

Focus on fixes on telecons

Ranges (P2836)

Deprecation (P2863)

Focus on big papers at F2F meetings

- mdspan (P1684)
- Unicode (P2728, P2729)
- SIMD (P2664, P2663, P2876)
- Policy (P1656, P2148)

## Core (Maurer)

We reviewed all the papers that were in our queue. A lot of them are on the straw polls, including some that we have already approved in earlier meetings, but couldn't move because we were busy with C++23.

We forwarded a number of papers with questions to EWG. When we notice issues in papers we alert EWG to decide if the situation is intentional or not.

Thank you to the minute takers.

Regarding P2620 (Improve the wording for Universal Character Names in identifiers), during review, we noticed a few changes in behavior and the author has expressed that he does not want to pursue that paper any more.

We faced some obstacles; when there were audio troubles in another room, CWG was without a chair. We need an assistant chair in core. Please send applications to me or Herb.

Poll 9, P2752, permits implementations to store the backing array of `std::initializer_list` in static storage, as opposed to on the stack, in certain restricted circumstances. That is possible visible in terms of optimisation potential

With P2169, you can use an `_` when declaring variables or structured bindings if you do not intend to refer to that thing afterwards. There was a lot of effort in making it backwards compatible.

Paper in poll 10 will let you generate a compile time message if your `static_assert` fails.

We also looked at some issues. We plan to have teleconferences.

## CWG polls

**1. Accept as a Defect Report and apply the proposed resolution of all issues except issues 2518, 2521, 2659, 2674, 2678, and 2691 in [P2796R0](#) (Core Language Working Group "ready" Issues for the February, 2023 meeting) to the C++ Working Paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**1. Accept as Defect Reports and apply the proposed resolutions of all issues in [P2922R0](#) (Core Language Working Group "ready" Issues for the June, 2023 meeting) to the C++ Working Paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**2. Accept as a Defect Report and apply the changes in [P2621R2](#) (UB? In my Lexer?) to the C++26 Working Paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**3. Accept as a Defect Report and apply the changes in [P1854R4](#) (Making non-encodable string literals ill-formed) to the C++26 Working Paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**4. Apply the changes in [P2361R6](#) (Unevaluated strings) to the C++26 Working Paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**5. Apply the changes in [P2558R2](#) (Add @, \$, and ` to the basic character set) to the C++26 Working Paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**6. Apply the changes in [P2738R1](#) (constexpr cast from void\*: towards constexpr type-erasure) to the C++26 Working Paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**7. Accept as a Defect Report and apply the changes in [P2915R0](#) (Proposed resolution for CWG1223) to the C++26 Working Paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**8. Accept as a Defect Report and apply the changes in [P2552R3](#) (On the ignorability of standard attributes) to the C++26 Working Paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**9. Accept as a Defect Report and apply the changes in [P2752R3](#) (Static storage for braced initializers) to the C++26 Working Paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**10. Apply the changes in [P2741R3](#) (User-generated static\_assert messages) to the C++26 Working Paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**11. Apply the changes in [P2169R4](#) (A nice placeholder with no name) to the C++26 Working Paper.**

No discussion.  
No objection to unanimous consent.

Motion passes.

## Library (Wakely)

We met all week. We worked through our queue of papers. We got rid of our backlog and would like to keep it gone.

Thank you to Dietmar for scribing

Most polls were reviewed before this meeting, but were waiting for a C++26 WP. Almost everything we saw at this meeting is in the polls. Some papers needed more work or came in too late on Friday.

Larger papers we saw this week include execution, and submdspan. Thank you to the authors and reviewers for their patience.

Submdspan is done and is in the polls today. It is extending the work done for mdspan in C++23.

We will continue the review of `std::execution` (P2300).

There were some questions raised about atomic min max paper (P0493) and its handling of floating point numbers and particular values. My recommendation is that we fix those issues later if we need to.

## LWG polls

**1. Apply the changes for all Tentatively Ready issues in [P2910R0](#) (C++ Standard Library Issues to be moved in Varna, Jun. 2023) to the C++ working paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**2. Apply the changes in [P2497R0](#) (Testing for success or failure of `<charconv>` functions) to the C++ working paper.**

Jonathan Wakely : this was supposed to be voted on in Issaquah, but it was missed from the poll page.

No objection to unanimous consent.

Motion passes.

**3. Apply the changes in [P2592R3](#) (Hashing support for `std::chrono` value classes) to the C++ working paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**4. Apply the changes in [P2587R3](#) (`to_string` or not `to_string`) to the C++ working paper.**

No discussion.

No objection to unanimous consent.  
Motion passes.

**5. Apply the changes in [P2562R1](#) (constexpr Stable Sorting) to the C++ working paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**6. Apply the changes in [P2545R4](#) (Read-Copy Update (RCU)) to the C++ working paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**7. Apply the changes in [P2530R3](#) (Hazard Pointers for C++26) to the C++ working paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**8. Apply the changes in [P2538R1](#) (ADL-proof std::projected) to the C++ working paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**9. Apply the changes in [P2495R3](#) (Interfacing stringstream with string\_view) to the C++ working paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**10. Apply the changes in [P2510R3](#) (Formatting pointers) to the C++ working paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**11. Apply the changes in [P2198R7](#) (Freestanding Feature-Test Macros and Implementation-Defined Extensions) to the C++ working paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**12. Apply the changes in [P2338R4](#) (Freestanding Library: Character primitives and the C library) to the C++ working paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**13. Apply the changes in [P2013R5](#) (Freestanding Language: Optional ::operator new) to the C++ working paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

~~**14. Apply the changes in [P0493R4](#) (Atomic maximum/minimum) to the C++ working paper.**~~

JF : as discussed on the reflector, there are issues with floating point min and max when it comes to handling NaN and -0/+0. We haven't considered this in the discussions. We think the poll should be rejected for now and we should come back with the paper after these issues have been considered.

John Spicer : has this been considered in LWG ?  
Jonathan Wakely : no.  
Herb Sutter : Jonathan, are you happy recalling this poll ?  
Jonathan Wakely : yes.

Poll has been recalled.

**15. Apply the changes in [P2363R5](#) (Extending associative containers with the remaining heterogeneous overloads) to the C++ working paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**16. Apply the changes in [P1901R2](#) (Enabling the Use of weak\_ptr as Keys in Unordered Associative Containers) to the C++ working paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**17. Apply the changes in [P1885R12](#) (Naming Text Encodings to Demystify Them) to the C++ working paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**18. Apply the changes in [P0792R14](#) (function\_ref: a type-erased callable reference) to the C++ working paper.**

No discussion.

Objections in the room.

In favour : 59 (40 in person + 19 online)

Opposed : 1 (1 in person + 0 online)

Abstain : 29 (22 in person + 7 online)

Consensus.

Passes.

**19 Apply the changes in [P2874R2](#) (Mandating Annex D) to the C++ working paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**20 Apply the changes in [P2757R3](#) (Type checking format args) to the C++ working paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**21 Apply the changes in [P2637R3](#) (Member visit) to the C++ working paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**22 Apply the changes in [P2641R4](#) (Checking if a union alternative is active) to the C++ working paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**23 Apply the changes in [P1759R6](#)  (Native handles and file streams) to the C++ working paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**24 Apply the changes in [P2697R1](#) (Interfacing bitset with string\_view) to the C++ working paper.**

No discussion.

No objection to unanimous consent.

Motion passes.

**25 Apply the changes in [P1383R2](#) (More constexpr for cmath and complex) to the C++ working paper.**

Jonathan Wakely : we forgot to bump the relevant feature test macro. I considered revising the poll to ask the editor to bump the macro, but we didn't discuss it in the group. It can be fixed with an issue. It doesn't need to stop the feature.

No discussion.  
Objections in the room.

In favour : 62 (42 in person + 20 online)  
Opposed : 2 (2 in person + 0 online)  
Abstain : 21 (13 in person + 8 online)

Consensus.  
Passes.

**26 Apply the changes in [P2734R0](#) (Adding the new 2022 SI prefixes) to the C++ working paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

**27 Apply the changes in [P2548R6](#) (copyable\_function) to the C++ working paper.**

No discussion.  
Objections in the room.

In favour : 50 (32 in person + 18 online)  
Opposed : 3 (3 in person + 0 online)  
Abstain : 31 (21 in person + 10 online)

Consensus.  
Passes.

**28 Apply the changes in [P2714R1](#) (Bind front and back to NTTP callables) to the C++ working paper.**

No discussion.  
Objections in the room.

In favour : 29 (20 in person + 7 online)  
Opposed : 1 (1 in person + 0 online)  
Abstain : 39 (27 in person + 12 online)

Consensus.  
Passes.

**29 Apply the changes in [P2630R4](#) (submdspan) to the C++ working paper.**

No discussion.  
No objection to unanimous consent.  
Motion passes.

## Direction Group (Stroustrup)

DG didn't meet in Varna. Continues biweekly meetings.

## WG21 polls

**1 Approve [P1000R5](#) as the schedule for C++26.**

Herb Sutter ; Same poll as what we took in Prague for C++23, updated for C++26. It adds 3 years to the schedule we had for C++23.

Peter Brett : do we expect for iso to publish C++23 in 2023?

Herb : maybe. Whatever date they put on it, it is C++23 in the marketplace. I am optimistic they will finish it by the end of the year.

No objection to unanimous consent.  
Motion passes.

## 8. Closing activities

### 8.1 Issues delayed until today

No discussion.

### 8.2 Mailings

Note: These are the closest regular mailings and not special pre/post meeting mailings.

2023-07-15: Post-Varna  
2024-10-15: Pre-Kona

## 9. INCITS/C++ Agenda Items

## 9.1 INCITS C++ Motions, if any

INCITS/C++ business.

[Note : during the INCITS motion, a comment was made about the possible need for a resolution meeting for “no” votes. After the plenary, John Spicer clarified that resolution meetings are never required for votes taken at a meeting, only those by letter ballot.]

### 9.1.3 Plans for the future (INCITS C++)

No discussion.

### 9.1.4 Next and following meetings

Herb Sutter : thank you to the host. This was the first pandemic meeting delayed, and then delayed again. It is great that it happened and thank you for the efforts in making it happen.  
John Spicer : thank you to the host, it was a beautiful meeting.

Mark Zeren presents slides about the organization of the meeting and the people involved in making it happen.

Mark Zeren : Thank you very much for coming and we hope you enjoyed your stay.

Herb presents.

2023-11-6/11: Kona, HI, USA (N4936)

2024-03-18/23: Tokyo, Japan (N4946)

This meeting is hosted by Woven by Toyota. There is an updated invitation paper coming.

JF Bastien presents the current paper.

JF Bastien : the paper says the meeting will be hosted in our office building, but that has changed. We are likely going to change the location to a place that is 12 minutes walk from the office building. I will update in the next couple of weeks. I got a request to recommend hotels. There are a lot of hotels to choose from. Mandarin oriental is expensive. Figure out what you want in a hotel. I recommend Japanese hotels.

March is the cherry blossom season. It gets busy so book a hotel and flights early.

We're hosting a two day conference with 1 track and only 10 talks.

There are two airports in Tokyo, both are convenient. Take the high speed rail.  
Most people don't need a visa, but check your own circumstances to be sure.

John Spicer : When should we book our trip ?

JF Bastien : I recommend you start looking now.

Herb Sutter presents.

Meeting after Japan is tentatively scheduled for Stockholm. I reached out to other potential hosts. If you are interested in hosting next summer, please send me an email.

In autumn 2024, we will be in Poland.

There are some tentative hosts for after that.

## 10. Adjournment (INCITS C++ motion)

Inbal Levi presents photos from the meeting.

INCITS/C++ motion to adjourn.

WG21 Motion to adjourn.

No objection.

Meeting adjourned at 10:16 AM GMT+3.

## 11. Attendance

| Attendee                 | Affiliation |
|--------------------------|-------------|
| Adams, Michael           | SCC         |
| Adelstein Lelbach, Bryce | ANSI        |
| Alday, Juan              | ANSI        |
| Amini, Parsa             | ANSI        |
| Arkipova, Olga           | ANSI        |
| Arutyunyan, Ruslan       | ANSI        |
| Ažman, Gašper            | BSI         |
| Baker, Billy             | ANSI        |
| Baker, Lewis             | ANSI        |
| Ballman, Aaron           | ANSI        |
| Balog, Pal               | ANSI        |

|                                          |      |
|------------------------------------------|------|
| Bastien, Jean-Francois                   | SCC  |
| Bentley, Matthew                         | ANSI |
| Berne, Joshua                            | ANSI |
| Bi, Brian                                | ANSI |
| Birbacher, Frank                         | ANSI |
| Boehm, Hans                              | ANSI |
| Bonaventura, Xavier                      | DIN  |
| Boric, Nemanja                           | ANSI |
| Boyarinov, Konstantin                    | ANSI |
| Brett, Peter                             | BSI  |
| Brown, Walter E                          |      |
| Butler, Matthew                          | ANSI |
| Büttner, Sebastian                       | ANSI |
| Cardoso de Souza Rodrigues,<br>Guilherme | ASI  |
| Catmur, Edward                           | BSI  |
| Caves, Jonathan                          | ANSI |
| Childers, Wyatt                          | ANSI |
| Chorazewicz, Igor                        | ANSI |
| Corden, Richard                          | ANSI |
| Craig, Benjamin                          | ANSI |
| Craig, Philip                            | BSI  |
| D'Angelo, Giuseppe                       | ANSI |
| Davidson, Guy                            | BSI  |
| de Wever, Mark                           | ANSI |
| Dimitrov, Georgi                         | BDS  |

|                             |       |
|-----------------------------|-------|
| Dimov, Peter                |       |
| Dominiak, Michał            | PKN   |
| Dos Reis, Gabriel           | AFNOR |
| Douglas, Niall              | NSAI  |
| Douglas, Robert             | ANSI  |
| Doumler, Timur              | BSI   |
| Downey, Steve               | ANSI  |
| Drewienkowski, Mateusz      |       |
| Dusikova, Hana              | UNMZ  |
| Engert, Daniela             | ANSI  |
| Erdogmus, Kaan              | ANSI  |
| Fertig, Andreas             | DIN   |
| Fevold, Jake                | ANSI  |
| Fevold, Jake                | ANSI  |
| Foco, Marco                 | UNI   |
| Fracassi, Fabio             | DIN   |
| Friedman, Ronen             | SII?  |
| García Sánchez, José Daniel | UNE   |
| Garland, Jeff               | ANSI  |
| Gill, Mungo                 | NSAI  |
| Gillard, Mark               | SFS   |
| Giroux, Olivier             | ANSI  |
| Goldblatt, David            | ANSI  |
| Goodspeed, Nathaniel        | ANSI  |
| Gordon, Fraser              | SCC   |
| Gruber, Bernhard Manfred    | SNV   |

|                        |       |
|------------------------|-------|
| Gustafsson, Bengt      | SIS   |
| Hagins, Jody           | ANSI  |
| Halpern, Pablo         | ANSI  |
| Hauswedell, Hannes     | IST   |
| Hava, Michael Florian  | ASI   |
| Hoemmen, Mark          | ANSI  |
| Hollman, Daisy         | ANSI  |
| Honermann, Tom         | ANSI  |
| Hunt, Oliver           | ANSI  |
| Jabot, Corentin        | AFNOR |
| Jaroszyńska, Magdalena | PKN   |
| Jevnik, Joseph         | ANSI  |
| Joly, Loïc             | AFNOR |
| Josuttis, Nicolai      | DIN   |
| KAMINISKI, Tomasz      | AFNOR |
| Keane, Erich           | ANSI  |
| Keremidchiev, Vassil   | BDS   |
| Khlebnikov, Rostislav  | ANSI  |
| Kirkovski, Seraphime   |       |
| Koepppe, Thomas        | ANSI  |
| Kokula, Volker         |       |
| Kosunen, Elias         | SFS   |
| Kretz, Matthias        | DIN   |
| Krügler, Daniel        | ANSI  |
| Krzemiński, Andrzej    | PKN   |
| Kuhl, Dietmar          | ANSI  |

|                        |      |
|------------------------|------|
| Kulczycki, Peter       | ASI  |
| Laine, Timothy         | ANSI |
| Lakos, John            | ANSI |
| Lakos, John            | ANSI |
| Leahy, Robert          | ANSI |
| Lebrun-Grandie, Damien | ANSI |
| Levi, Inbal            | SII  |
| Liber, Nevin           | ANSI |
| Lopes, Bruno           | ANSI |
| Lukanov, Stanimir      | BDS  |
| Machutova, Jana        | UNMZ |
| MacLean, Colin         | ANSI |
| Maness, Wesley         | ANSI |
| Marsman, A.            | NEN  |
| Maurer, Jens           | ANSI |
| Meerwald, Christof     | ASI  |
| Meredith, Alisdair     | ANSI |
| Merrill, Jason         | ANSI |
| Mihaylov, Mihail       | BDS  |
| Miller, William        | ANSI |
| Morales, Nicolas       | ANSI |
| Moschovakos, Paris     | SNV  |
| Müller, Jonathan       | DIN  |
| Murzin, Sergei         | ANSI |
| Neatu, Darius          | ANSI |
| Neumann, Thomas        |      |

|                               |        |
|-------------------------------|--------|
| Niebler, Eric                 | ANSI   |
| Nishanov, Gor                 | ANSI   |
| O'Dwyer, Arthur               | ANSI   |
| Opara, Jolanta                | PKN    |
| Orr, Roger                    | BSI    |
| Owen, Nathan                  | ANSI   |
| Persson, Jonas                | SIS    |
| Polukhin, Anton               | GOST R |
| Preney, Paul                  | SCC    |
| Pusz, Mateusz                 | ANSI   |
| Ranns, Nina Dinka             | BSI    |
| Reverdy, Vincent              | AFNOR  |
| Revzin, Barry                 | ANSI   |
| Rigault, Jean-Paul            | AFNOR  |
| Rivera Morell, René Ferdinand | ANSI   |
| Rosten, Oliver                | BSI    |
| Roy, Patrice                  | SCC    |
| Ruoso, Daniel                 | ANSI   |
| Ryan, Christopher             | ANSI   |
| Sandoe, Iain                  | BSI    |
| Sankel, David                 | ANSI   |
| Sattler, Florian              | ANSI   |
| Schurr, Stephen               | ANSI   |
| Scogland, Thomas              | ANSI   |
| Seymour, William              | ANSI   |
| Simpson, Robert               | ANSI   |

|                     |                     |
|---------------------|---------------------|
| Sitnikov, Sasha     |                     |
| Snyder, Jeff        | BSI                 |
| Song, Tim           | ANSI                |
| Spencer, Michael    | ANSI                |
| Spertus, Michael    | ANSI                |
| Spicer, John        | ANSI                |
| St. Amour, Bryan    | SCC                 |
| Stroustrup, Bjarne  | ANSI                |
| Sutter, Herb        | ISO/IEC JTC 1/SC 22 |
| Tenty, David        | ANSI                |
| Tomazos, Andrew     |                     |
| Tong, Hubert        | SCC                 |
| van Winkel, J.C.    | NEN                 |
| Vandevoorde, Daveed | ANSI                |
| Vasama, Lauri       | SFS                 |
| Vasilev, Vasil      | BDS                 |
| Vitek, David        | ANSI                |
| Voicu, Alexandru    | ANSI                |
| Vollmann, Detlef    | SNV                 |
| Vormwald, Steven    | ANSI                |
| Voss, Michael       | ANSI                |
| Voutilainen, Ville  | SFS                 |
| Vrankar, Domen      |                     |
| Wakely, Jonathan    | ANSI                |
| Walker, Kelly       | ANSI                |
| Waterloo, Jarrad    | ANSI                |

|                   |      |
|-------------------|------|
| Weis, Andreas     | DIN  |
| Williams, Anthony | BSI  |
| Williamson, Jerry |      |
| Wong, Michael     | SCC  |
| Xie, Hui          | BSI  |
| xu, chuanqi       | SAC  |
| Yaghmour, Shafik  | ANSI |
| Yonchev, Zamfir   |      |
| Yuan, Zhihao      | ANSI |
| Zeren, Mark       | ANSI |
| Zhdan, Egor       |      |
| Zielonka, Bartosz | PKN  |
| Zissu, Andrei     | SII  |
| Zverovich, Victor | ANSI |